

Advanced (Habanero)

Q1. Rationalize the denominator of $\frac{1}{\sqrt{2}}$.

- (A) $\frac{\sqrt{2}}{2}$
(B) $\frac{1}{2\sqrt{2}}$
(C) $\frac{2}{\sqrt{2}}$
(D) $\frac{\sqrt{2}}{4}$
(E) $\frac{1}{4\sqrt{2}}$

Q2. What is the rationalized form of $\frac{1}{\sqrt{5}-1}$?

- (A) $\frac{\sqrt{5}+1}{4}$
(B) $\frac{\sqrt{5}-1}{4}$
(C) $\frac{5+1}{\sqrt{5}-1}$
(D) $\frac{\sqrt{5}+1}{5-1}$
(E) $\frac{5-1}{\sqrt{5}+1}$

Q3. Simplify $\frac{2}{\sqrt{7}+\sqrt{3}}$ by rationalizing the denominator.

- (A) $\frac{2(\sqrt{7}-\sqrt{3})}{4}$
(B) $\frac{2(\sqrt{7}+\sqrt{3})}{4}$
(C) $\frac{\sqrt{7}+\sqrt{3}}{2}$
(D) $\frac{2(\sqrt{7}-\sqrt{3})}{7-3}$
(E) $\frac{2}{\sqrt{7}-\sqrt{3}}$

Q4. Rationalize the denominator of $\frac{\sqrt{3}}{\sqrt{6}+\sqrt{2}}$.

- (A) $\frac{\sqrt{18}-\sqrt{6}}{4}$
(B) $\frac{\sqrt{18}+\sqrt{6}}{4}$
(C) $\frac{\sqrt{6}+\sqrt{2}}{\sqrt{3}}$
(D) $\frac{\sqrt{3}(\sqrt{6}-\sqrt{2})}{4}$
(E) $\frac{\sqrt{6}-\sqrt{2}}{\sqrt{3}}$

Q5. Simplify $\frac{\sqrt{2}+1}{\sqrt{5}-1}$ by rationalizing the denominator.

- (A) $\frac{\sqrt{10}+\sqrt{2}+5+1}{4}$
(B) $\frac{\sqrt{10}-\sqrt{2}+5-1}{4}$
(C) $\frac{\sqrt{10}+\sqrt{2}-5-1}{4}$
(D) $\frac{\sqrt{2}+1}{\sqrt{5}+1}$
(E) $\frac{\sqrt{10}-\sqrt{2}-5+1}{4}$

Q6. Rationalize the denominator of $\frac{1}{\sqrt{8}-2}$.

- (A) $\frac{\sqrt{8}+2}{4}$
(B) $\frac{\sqrt{8}-2}{4}$
(C) $\frac{1}{\sqrt{8}+2}$
(D) $\frac{\sqrt{8}+2}{4}$
(E) $\frac{2+\sqrt{8}}{4}$

Q7. Simplify $\frac{\sqrt{6}}{\sqrt{3}+1}$ by rationalizing the denominator.

- (A) $\frac{\sqrt{6}-\sqrt{2}}{2}$
(B) $\frac{\sqrt{6}+\sqrt{2}}{2}$
(C) $\frac{\sqrt{3}+1}{\sqrt{6}}$
(D) $\frac{\sqrt{18}-\sqrt{6}}{2}$
(E) $\frac{\sqrt{2}+\sqrt{6}}{2}$

Q8. Rationalize the denominator of $\frac{1}{\sqrt{12}-\sqrt{3}}$.

- (A) $\frac{\sqrt{12}+\sqrt{3}}{9}$
(B) $\frac{\sqrt{12}-\sqrt{3}}{9}$
(C) $\frac{\sqrt{3}+\sqrt{12}}{3}$
(D) $\frac{\sqrt{12}+\sqrt{3}}{3}$
(E) $\frac{1}{\sqrt{12}+\sqrt{3}}$

Open Ended Questions (FRQ)

Q9. Rationalize the denominator of $\frac{\sqrt{x}}{\sqrt{y}+\sqrt{z}}$. Show all steps.

Q10. Simplify $\frac{\sqrt{a}+\sqrt{b}}{\sqrt{c}-\sqrt{d}}$ by rationalizing the denominator. Assume a, b, c, d are positive real numbers.

Chef's Tips

- Provide clear and concise instructions.
- Encourage students to show their work and reasoning.
- Remind students to rationalize the denominator in the required problems.